



**MID SEMESTER EXAMINATION, AUTUMN-2024**  
School of Applied Sciences, KIIT-DU

B. Tech.  
1<sup>st</sup> Semester (2024AB)  
Autumn 2024 (SAS)

|                                                   |               |                |                  |
|---------------------------------------------------|---------------|----------------|------------------|
| Course: Differential Equations and Linear Algebra | Code: MA11001 | Full Marks: 20 | Time: 90 minutes |
|---------------------------------------------------|---------------|----------------|------------------|

Answer any **FOUR** QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

| Q.1 | Answer the following questions                                                                                         | Marks | CO  |
|-----|------------------------------------------------------------------------------------------------------------------------|-------|-----|
| a)  | Solve $\tan y \frac{dy}{dx} = \sin(x+y) + \sin(x-y)$ .                                                                 | 1     | CO1 |
| b)  | Form the differential equation by eliminating the arbitrary constants:<br>$y = Ax + A^2$ .                             | 1     | CO1 |
| c)  | Find the value of $\mu$ for which the differential equation :<br>$(xy^2 + \mu x^2y)dx + (x+y)x^2dy = 0$ is exact.      | 1     | CO1 |
| d)  | For what value of $a$ ; $\ln x$ and $\ln x^a$ are linearly independent.                                                | 1     | CO2 |
| e)  | Find the order and degree of the differential equation $y = \sqrt{x} \frac{dy}{dx} + \left( \frac{dy}{dx} \right)^k$ . | 1     | CO1 |

| Q.2 |                                                                                                          | Marks | CO  |
|-----|----------------------------------------------------------------------------------------------------------|-------|-----|
| a)  | Solve the differential equation $(2xy + x^2)y' = 3y^2 + 2xy$ .                                           | 3     | CO1 |
| b)  | Test the exactness and solve the differential equation given by<br>$(2xydx + dy)e^{x^2} = 0, y(0) = 2$ . | 2     | CO1 |

| Q.3 |                                                                                              | Marks | CO  |
|-----|----------------------------------------------------------------------------------------------|-------|-----|
| a)  | Solve the linear differential equation $x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$ . | 3     | CO1 |
| b)  | Find the orthogonal trajectory of the family of curves : $x = ce^{-y^2}$ ..                  | 2     | CO1 |

| Q.4 |                                                                                                                                                                                                                          | Marks | CO  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| a)  | If in a culture of yeast, the rate of growth is proportional to the amount $y(t)$ present at time $t$ , and if $y(t)$ doubles in 1 day, how much can be expected after 3 days at the same rate of growth? After 1 week ? | 3     | CO1 |
| b)  | Solve the non-linear equation $y'' + y'^3 \cos y = 0$ .                                                                                                                                                                  | 2     | CO1 |

| Q.5 |                                                                                                     | Marks | CO  |
|-----|-----------------------------------------------------------------------------------------------------|-------|-----|
| a)  | Solve the Bernoulli's equation $xy' + y = y^2 \log x$ .                                             | 3     | CO1 |
| b)  | Reduce to first order and solve the differential equation:<br>$x^2y'' - 5xy' + 9y = 0, y_1 = x^3$ . | 2     | CO2 |

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